Random Variables and Distribution Functions

Probability Theory and Mathematical Statistics

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Chapter 2, Section 1

Outline

1. Concept of a Random Variable
2. Distribution Function
3. Basic Properties of the Distribution Function
4. Summary and Exercises

From Functions to Random Variables

In calculus, a function is a rule that assigns to each real number x a unique real number $y = f(x)$.

In probability theory, a random variable is also a function, but:

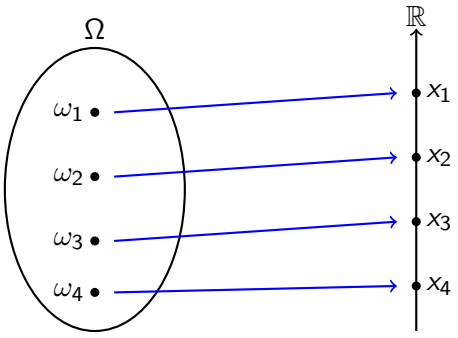
- Input: a sample point ω in the sample space Ω;
- Output: a real number $X(\omega)$.

Key idea

A random variable is a **real-valued function defined on the sample space**:

$$X : \Omega \to \mathbb{R}.$$

Illustration: Random Variable as a Mapping



Interpretation

Each sample point ω is mapped to a real number $X(\omega)$.

Definition of a Random Variable

Definition 1.1 (Random Variable)

Let X be a function on the sample space Ω with values in $\mathbb{R}$, i.e., $X : \Omega \to \mathbb{R}$. If for every real x the preimage

$$X^{-1}\big((-\infty, x]\big) = \{\omega \in \Omega : X(\omega) \leq x\}$$

is an event, i.e., $\{X \leq x\}$ is an event, then X is called a **random variable**.

Key point

A random variable is essentially a **real-valued function on the sample space**. Its input is a sample point (which may not be a number) and its output is a real number.

Remark (Random Variable via Inverse Mapping)

Remark 1.1

More generally and rigorously, let $X : \Omega \to \mathbb{R}$ and require that for every real set $B \subset \mathbb{R}$, its preimage

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$$

is an event (i.e., belongs to the event field $\mathcal{F}$). Such a function X is called a random variable.

Why Only Intervals $(-\infty, x]$ Are Required

Explanation

In the formal definition, we only require that the preimage of intervals of the form $(-\infty, x]$ is an event. Why is this enough?

- Any subset of $\mathbb{R}$ can be obtained from intervals $(-\infty, x]$ by finitely or countably many operations of intersection, union, and complement.
- The event field $\mathcal{F}$ is closed under these operations.
- Therefore, requiring $\{X \leq x\}$ to be an event for every real x already guarantees that $X^{-1}(B)$ is an event for every real set B.

In other words, the simplified requirement "$\{X \leq x\}$ is an event" already contains all the information needed to say that "X is a random variable."

Example 1.1 (Statement)

Example 1.1 (Constant is a Random Variable)

Let Ω be any sample space and c a fixed real number. Define

$$X(\omega) = c, \qquad \omega \in \Omega.$$

Prove that X is a random variable using the inverse mapping method.

Example 1.1 (Solution)

Solution

For any real number x, consider the preimage

$$X^{-1}\big((-\infty, x]\big) = \{\omega \in \Omega : X(\omega) \leq x\}.$$

Since $X(\omega) = c$ for every ω,

$$X(\omega) \leq x \iff c \leq x.$$

Therefore,

$$X^{-1}\big((-\infty, x]\big) = \begin{cases} \varnothing, & x < c, \\ \Omega, & x \geq c. \end{cases}$$

Both $\varnothing$ and Ω are events, so X is a random variable.

Remark 1.2 (Constant as a Degenerate Random Variable)

Remark 1.2

This example shows that a **constant can be viewed as a degenerate case of a random variable**. Although in practice we usually do not regard a constant as a random variable, by definition it fully satisfies "for every sample point ω, there is a uniquely determined real number $X(\omega)$ assigned."

Interpretation via inverse mapping

The constant $X \equiv c$ "presses" the entire sample space Ω onto a single point c on the real line. For any x, the preimage is either $\varnothing$ (when $x < c$) or Ω (when $x \geq c$), both of which are legitimate events. Hence X satisfies the definition of a random variable.

Example 1.2 (Statement)

Example 1.2

Toss a coin with sample space $S = \{H, T\}$. Define

$$X(H) = 1, \qquad X(T) = 0.$$

Show that X is a random variable, and identify what it represents.

Example 1.2 (Solution)

Solution

For any real number x, the preimage

$$X^{-1}\big((-\infty, x]\big) = \{\omega \in S : X(\omega) \leq x\}$$

is one of the following:

- $\varnothing$, if $x < 0$;
- $\{T\}$, if $0 \leq x < 1$;
- $S = \{H, T\}$, if $x \geq 1$.

All of these are events. Hence X is a random variable.
It represents the **number of heads** in one toss.

Example 1.3 (Statement)

Example 1.3

Roll a die with sample space $S = \{1, 2, 3, 4, 5, 6\}$. Define

$$X(\omega) = \omega.$$

Show that X is a random variable and describe its meaning.

Example 1.3 (Solution)

Solution

For any real number x, the preimage

$$X^{-1}\big((-\infty, x]\big) = \{\omega \in S : \omega \leq x\}$$

is a subset of S, hence an event. Thus X is a random variable.
It represents the **number shown on the die**.

Comparison with Functions in Calculus

	Function in calculus	**Random variable**
Domain	Real set D	Sample space Ω
Argument	Real number x	Sample point ω
Value	Real number $y = f(x)$	Real number $X = X(\omega)$
Rule	Given x, output $f(x)$	Given ω, output $X(\omega)$

Key point

A random variable and a function in calculus have exactly the same structure; the only difference is the **domain**: a calculus function is defined on a real set, while a random variable is defined on the sample space.

Definition of the Distribution Function

Definition 1.2 (Distribution Function)

Let X be a random variable. For any real number x, the function

$$F(x) = P(X \leq x)$$

is called the **distribution function** of X. We say X follows $F(x)$ and write $X \sim F(x)$.

Meaning

$F(x)$ is the probability of the event $\{X \leq x\}$. Since $\{X \leq x\}$ is a subset of the sample space, its probability is determined, so $F(x)$ is a function on $\mathbb{R}$ with values in $[0, 1]$.

Using the CDF to Compute All Probabilities

Key fact

Once the distribution function $F(x) = P(X \leq x)$ is known, the probability of **any** event involving X can be computed from F.

Basic formulas

For any real numbers $a < b$:

$$
\begin{aligned}
P(a < X \leq b) &= F(b) - F(a), \\
P(X > a) &= 1 - F(a), \\
P(X < b) &= F(b - 0), \\
P(X = a) &= F(a) - F(a - 0),
\end{aligned}
$$

where $F(b - 0) = \lim_{x \to b^-} F(x)$ is the left-hand limit of F at b.

Proof of the Basic Formulas

Proof of $P(a < X \leq b) = F(b) - F(a)$

Write the event $\{a < X \leq b\}$ as a difference:

$$\{a < X \leq b\} = \{X \leq b\} \setminus \{X \leq a\}.$$

Since $\{X \leq a\} \subset \{X \leq b\}$, by the difference rule for probability,

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a).$$

Proof of $P(X > a) = 1 - F(a)$

The events $\{X > a\}$ and $\{X \leq a\}$ are complementary:

$$\{X > a\} \cup \{X \leq a\} = \Omega, \qquad \{X > a\} \cap \{X \leq a\} = \varnothing.$$

Hence

$$P(X > a) = 1 - P(X \leq a) = 1 - F(a).$$

Proof of the Basic Formulas (continued)

Proof of $P(X < b) = F(b - 0)$

Let $x_n \uparrow b$. Then $\{X \leq x_n\} \uparrow \{X < b\}$. By the continuity of probability from below,

$$P(X < b) = \lim_{n \to \infty} P(X \leq x_n) = \lim_{x \to b^-} F(x) = F(b - 0).$$

Proof of $P(X = a) = F(a) - F(a - 0)$

Write

$$\{X = a\} = \{X \leq a\} \setminus \{X < a\}.$$

Since $\{X < a\} \subset \{X \leq a\}$,

$$P(X = a) = P(X \leq a) - P(X < a) = F(a) - F(a - 0).$$

□

Why the CDF Determines Everything

From intervals to all events

The formulas above handle all intervals of the form $(a, b]$, $(-\infty, b)$, $(a, +\infty)$, and singletons $\{a\}$.

Any event is a countable combination of intervals

Every event involving X can be expressed as a countable union, intersection, or complement of such intervals. For example:

- $\{X \in (a, b)\} = \bigcup_{n=1}^{\infty} \left\{a + \frac{1}{n} < X \leq b - \frac{1}{n}\right\}$;
- $\{X \leq a\}$ and $\{X > a\}$ are already intervals.

By the countable additivity of probability, once the probabilities of all intervals are known, the probability of any event is determined.

Conclusion

The distribution function $F(x) = P(X \leq x)$ **completely describes the probability law of** X. Two random variables with the same CDF have the same probability distribution.

Example 1.4 (Statement)

Example 1.4 (Degenerate Distribution)

Let the random variable $X \equiv c$, i.e., X takes the constant value c with probability 1. Find the distribution function $F(x)$.

Example 1.4 (Solution)

Solution

By the definition of the distribution function,

$$F(x) = P(X \leq x).$$

Since $X \equiv c$, the event $\{X \leq x\}$ depends on the comparison between x and c:

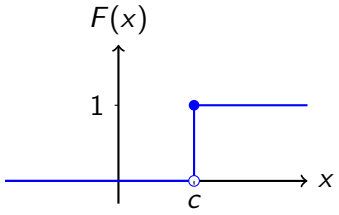
- If $x < c$: $X \leq x$ cannot occur, i.e., $\{X \leq x\} = \varnothing$, so $F(x) = 0$;
- If $x \geq c$: $X \leq x$ is certain, i.e., $\{X \leq x\} = \Omega$, so $F(x) = 1$.

Therefore,

$$F(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c. \end{cases}$$

This is a **step function** with a single jump at $x = c$ of height 1. This distribution is called the **degenerate distribution** and is denoted $X \equiv c$.

Example 1.4 (Graph)



Example 1.5 (Statement)

Example 1.5 (Discrete Distribution: Two Values)

Toss a fair coin. Let

$$X = \begin{cases} 1, & \text{if heads,} \\ 0, & \text{if tails.} \end{cases}$$

Find the distribution function $F(x)$.

Example 1.5 (Solution)

Solution

X takes values 0 and 1, with

$$P(X = 0) = \frac{1}{2}, \qquad P(X = 1) = \frac{1}{2}.$$

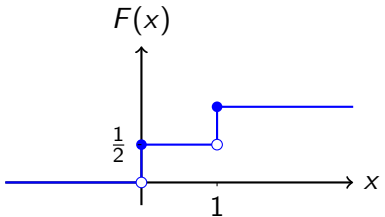
For any real x:

- If $x < 0$: $\{X \leq x\} = \varnothing$, so $F(x) = 0$;
- If $0 \leq x < 1$: $\{X \leq x\} = \{X = 0\}$, so $F(x) = \frac{1}{2}$;
- If $x \geq 1$: $\{X \leq x\} = S$, so $F(x) = 1$.

Hence

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{2}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Example 1.5 (Graph)



Example 1.6 (Statement)

Example 1.6 (Continuous Type: Random Number in an Interval)

Randomly pick a number X from the interval $[0, 1]$, with all points equally likely. Find the distribution function $F(x)$.

Example 1.6 (Solution, Part 1)

Solution

By definition,

$$F(x) = P(X \leq x).$$

We discuss three cases depending on the position of x relative to $[0, 1]$.
Case 1: $x < 0$**.** The event $\{X \leq x\}$ cannot occur, i.e., $\{X \leq x\} = \varnothing$, so

$$F(x) = P(\varnothing) = 0.$$

Example 1.6 (Solution, Part 2)

Solution (continued)

Case 2: $0 \leq x \leq 1$**.** The event $\{X \leq x\}$ means "X falls in $[0, x]$". Since X is equally likely in $[0, 1]$, the probability is the ratio of lengths:

$$F(x) = P(X \leq x) = \frac{\text{length of } [0, x]}{\text{length of } [0, 1]} = \frac{x}{1} = x.$$

Case 3: $x > 1$**.** The event $\{X \leq x\}$ is certain, i.e., $\{X \leq x\} = \Omega$, so

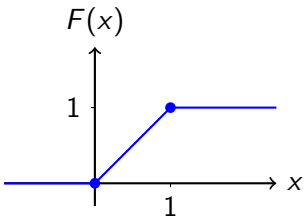
$$F(x) = P(\Omega) = 1.$$

Example 1.6 (Result and Graph)

Result

$$F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

This distribution function is **continuous** on the whole real line, with no jumps.



Comparison: Discrete vs. Continuous

Remark 1.3

Comparing the two distribution functions:

- **Discrete type (coin toss)**: the distribution function is a step function, with jumps at $x = 0$ and $x = 1$;
- **Continuous type (random number in an interval)**: the distribution function is continuous, with no jumps.

This reflects the essential difference in shape between discrete and continuous random variables.

Example 1.7 (Statement)

Example 1.7 (Mixed Type: Discrete + Continuous)

Consider the following experiment: first toss a fair coin.

- If the coin shows heads, set $X = 1$;
- If the coin shows tails, randomly pick a number from $[0, 1]$ with all points equally likely, and let X be this number.

Find the distribution function $F(x)$.

Example 1.7 (Setup)

Setup

Let

$$A = \{\text{coin shows heads}\}, \qquad A^c = \{\text{coin shows tails}\}.$$

Then

$$P(A) = P(A^c) = \frac{1}{2}.$$

Conditional distributions

When A occurs, $X = 1$ always, so

$$P(X \le x \mid A) = \begin{cases} 0, & x < 1, \\ 1, & x \ge 1. \end{cases}$$

When A^c occurs, X is uniformly distributed on $[0, 1]$, so

$$\begin{cases} 0, & x < 0, \end{cases}$$

Example 1.7 (Solution, Part 1)

Solution

By the law of total probability, for any real x,

$$F(x) = P(A)P(X \leq x \mid A) + P(A^c)P(X \leq x \mid A^c).$$

Case 1: $x < 0$**.** Both conditional probabilities are 0, so

$$F(x) = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 0 = 0.$$

Example 1.7 (Solution, Part 2)

Solution (continued)

Case 2: $0 \leq x < 1$. We have

$$P(X \leq x \mid A) = 0, \qquad P(X \leq x \mid A^c) = x,$$

so

$$F(x) = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot x = \frac{x}{2}.$$

Case 3: $x \geq 1$. Both conditional probabilities are 1, so

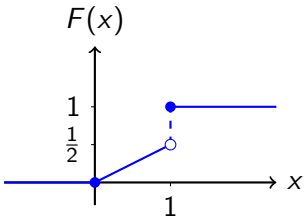
$$F(x) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1.$$

Example 1.7 (Result)

Result

$$F(x) = \begin{cases} 0, & x < 0, \\ \dfrac{x}{2}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Example 1.7 (Graph)



Remark 1.4 (Shape of the Mixed Distribution)

Remark 1.4

The distribution function obtained above has the following features:

- On $[0,1)$, $F(x) = \dfrac{x}{2}$ is **continuous and strictly increasing**; this part corresponds to "randomly pick a number from $[0,1]$."
- At $x = 1$, $F(x)$ jumps from $\frac{1}{2}$ to 1, with jump size

$$F(1) - F(1-0) = 1 - \frac{1}{2} = \frac{1}{2};$$

  this jump corresponds to the discrete part "set $X = 1$."

Hence this distribution function has both a continuous part and a jump: it is neither a pure discrete step function nor a pure continuous function, but a **mixed-type** distribution function.

Remark 1.5 (Mixing Discrete and Continuous)

Remark 1.5

This example shows that **mixing a discrete random variable and a continuous random variable with certain probabilities yields a mixed-type random variable**. Its distribution function can be obtained by the law of total probability: weight the two cases and add.

Theorem (Basic Properties of the Distribution Function)

Theorem 1.1

Let $F(x)$ be the distribution function of a random variable X. Then:

1. **Monotonicity**: $F(x)$ is nondecreasing on $\mathbb{R}$; that is, for any $x_1 < x_2$,

$$F(x_1) \leq F(x_2).$$

2. **Boundedness**: for any $x \in \mathbb{R}$, $0 \leq F(x) \leq 1$, and

$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1.$$

3. **Right-continuity**: $F(x)$ is right-continuous; that is, for any $x_0 \in \mathbb{R}$,

$$\lim_{x \to x_0^+} F(x) = F(x_0).$$

Proof of Monotonicity

Proof of (1) Monotonicity

Let $x_1 < x_2$. Consider the two events

$$A_1 = \{X \leq x_1\}, \qquad A_2 = \{X \leq x_2\}.$$

If $\omega \in A_1$, then $X(\omega) \leq x_1 < x_2$, so $\omega \in A_2$. Hence $A_1 \subset A_2$.
By the monotonicity of probability,

$$P(A_1) \leq P(A_2).$$

That is,

$$F(x_1) = P(X \leq x_1) \leq P(X \leq x_2) = F(x_2).$$

So $F(x)$ is nondecreasing on $\mathbb{R}$.

Proof of Boundedness (Part 1)

Proof: $0 \leq F(x) \leq 1$

Since $F(x) = P(X \leq x)$ is a probability, by the nonnegativity axiom,

$$F(x) \geq 0.$$

Since $\{X \leq x\} \subset \Omega$ and $P(\Omega) = 1$, by the monotonicity of probability,

$$F(x) = P(X \leq x) \leq P(\Omega) = 1.$$

Therefore,

$$0 \leq F(x) \leq 1.$$

Proof of Boundedness (Part 2)

Proof: $\lim_{x \to -\infty} F(x) = 0$

Let $x_n \downarrow -\infty$. Then the events

$$A_n = \{X \leq x_n\}$$

form a decreasing sequence:

$$A_1 \supset A_2 \supset \cdots$$

Moreover,

$$\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} \{X \leq x_n\} = \varnothing.$$

By the continuity of probability from above,

$$\lim_{n \to \infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = P(\varnothing) = 0.$$

Hence

Proof of Boundedness (Part 3a): Setting Up the Events

Proof: $\lim_{x \to +\infty} F(x) = 1$

Let $x_n \uparrow +\infty$. Then the events

$$B_n = \{X \leq x_n\}$$

form an increasing sequence:

$$B_1 \subset B_2 \subset \cdots$$

Moreover,

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} \{X \leq x_n\} = \Omega.$$

Indeed, for any $\omega \in \Omega$, $X(\omega)$ is a real number, so there exists N with $X(\omega) \leq x_N$, hence $\omega \in B_N$.

Proof of Boundedness (Part 3b): Taking the Limit

Proof (continued)

By the continuity of probability from below,

$$\lim_{n\to\infty} P(B_n) = P\left(\bigcup_{n=1}^{\infty} B_n\right) = P(\Omega) = 1.$$

Hence

$$\lim_{x\to+\infty} F(x) = 1.$$

Remark

Since F is nondecreasing, for any $x > n$ we have $F(n) \leq F(x) \leq 1$. Letting $n \to \infty$ gives the same conclusion.

Proof of Right-Continuity (Part 1)

Proof

Let $x_n \downarrow x_0$. Then the events

$$C_n = \{X \leq x_n\}$$

form a decreasing sequence:

$$C_1 \supset C_2 \supset \cdots$$

Moreover,

$$\bigcap_{n=1}^{\infty} C_n = \bigcap_{n=1}^{\infty} \{X \leq x_n\} = \{X \leq x_0\}.$$

Proof of Right-Continuity (Part 2)

Proof (continued)

Indeed:

- If $\omega \in \bigcap_n C_n$, then $X(\omega) \leq x_n$ for all n; since $x_n \downarrow x_0$, we get $X(\omega) \leq x_0$.
- Conversely, if $X(\omega) \leq x_0$, then $X(\omega) \leq x_n$ for all n (since $x_n > x_0$), so $\omega \in \bigcap_n C_n$.

By the continuity of probability from above,

$$\lim_{n\to\infty} P(C_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right) = P(X \leq x_0) = F(x_0).$$

Hence

$$\lim_{x\to x_0^+} F(x) = F(x_0).$$

□

Example 1.8 (Statement)

Example 1.8 (Checking Whether a Function is a CDF)

Determine whether each of the following functions is the distribution function of some random variable, and explain why:

(a) $F_1(x) = \dfrac{1}{1+e^x}, \qquad x \in \mathbb{R};$

(b) $F_2(x) = \begin{cases} 0, & x \le 0, \\ 1, & x > 0; \end{cases}$

(c) $F_3(x) = \begin{cases} 0, & x < 0, \\ \dfrac{1}{2}, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$

Example 1.8 (Solution, Part 1)

Solution (a)

Differentiate F_1:

$$F_1'(x) = \frac{-e^x}{(1+e^x)^2} < 0, \qquad x \in \mathbb{R}.$$

So F_1 is strictly decreasing, which violates monotonicity. Also,

$$\lim_{x \to -\infty} F_1(x) = 1, \qquad \lim_{x \to +\infty} F_1(x) = 0,$$

which also violates the required limits. Hence F_1 is **not** a CDF.

Example 1.8 (Solution, Part 2)

Solution (b)

Monotonicity holds: $F_2(x) = 0$ for $x \leq 0$ and $F_2(x) = 1$ for $x > 0$, so F_2 is nondecreasing.
Boundedness holds:

$$0 \leq F_2(x) \leq 1, \qquad \lim_{x \to -\infty} F_2(x) = 0, \qquad \lim_{x \to +\infty} F_2(x) = 1.$$

But right-continuity fails at $x = 0$:

$$F_2(0) = 0, \qquad \lim_{x \to 0^+} F_2(x) = 1.$$

So F_2 is **not** a CDF.

Example 1.8 (Solution, Part 3)

Solution (c)

Check the three properties one by one.
Monotonicity: F_3 takes values 0, $\frac{1}{2}$, 1 on $(-\infty, 0)$, $[0, 1)$, $[1, +\infty)$, so it is nondecreasing.
Boundedness:

$$0 \leq F_3(x) \leq 1, \qquad \lim_{x \to -\infty} F_3(x) = 0, \qquad \lim_{x \to +\infty} F_3(x) = 1.$$

Right-continuity: only need to check $x = 0$ and $x = 1$:

$$F_3(0) = \frac{1}{2} = \lim_{x \to 0^+} F_3(x), \qquad F_3(1) = 1 = \lim_{x \to 1^+} F_3(x).$$

Hence F_3 is right-continuous. Therefore, F_3 is a CDF.

Remark 1.6 (How to Check a CDF)

Remark 1.6

To determine whether a function is a CDF, check the three basic properties one by one:

1. **Monotonicity**: F is nondecreasing on $\mathbb{R}$;
2. **Boundedness**: $0 \leq F(x) \leq 1$, with $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$;
3. **Right-continuity**: F is right-continuous at every point.

All three are required:

- F_1 satisfies the limit conditions but fails monotonicity (strictly decreasing);
- F_2 satisfies monotonicity and boundedness but fails right-continuity;
- F_3 satisfies all three, so it is a CDF.

Example 1.9 (Statement)

Example 1.9 (Determining Constants from Boundedness)

Let the distribution function of a random variable X be

$$F(x) = A + B \arctan x, \qquad -\infty < x < +\infty.$$

Find:

1. the constants A and B;
2. $P\left(1 < X < \sqrt{3}\right)$.

Example 1.9 (Solution, Part 1)

Solution (1): Find A and B

By the boundedness property of the CDF,

$$\lim_{x\to-\infty} F(x) = 0, \qquad \lim_{x\to+\infty} F(x) = 1.$$

Since

$$\lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}, \qquad \lim_{x\to+\infty} \arctan x = \frac{\pi}{2},$$

we have

$$0 = A - \frac{\pi}{2}B, \qquad 1 = A + \frac{\pi}{2}B.$$

Adding gives $2A = 1$, so $A = \frac{1}{2}$. Subtracting gives $\pi B = 1$, so $B = \frac{1}{\pi}$.
Hence

$$F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x.$$

Example 1.9 (Solution, Part 2)

Solution (2): Compute $P(1 < X < \sqrt{3})$

By the properties of the CDF,

$$P(a < X \leq b) = F(b) - F(a).$$

Since $F(x)$ is continuous, $P(X = 1) = P(X = \sqrt{3}) = 0$, so

$$P(1 < X < \sqrt{3}) = P(1 < X \leq \sqrt{3}) = F(\sqrt{3}) - F(1).$$

Substitute $F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x$:

$$F(\sqrt{3}) = \frac{1}{2} + \frac{1}{\pi} \cdot \frac{\pi}{3} = \frac{5}{6},$$

$$F(1) = \frac{1}{2} + \frac{1}{\pi} \cdot \frac{\pi}{4} = \frac{3}{4}.$$

Therefore,

Remark 1.7 (Two Properties Used in Example 1.9)

Remark 1.7

This example uses two properties of the distribution function:

- **Boundedness**: from $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$ we determine A and B;
- **Interval probability**: from $P(a < X \leq b) = F(b) - F(a)$ we compute the probability of an interval.

Moreover, since $F(x)$ is continuous in this example, a continuous random variable assigns probability zero to any single point, so

$$P(1 < X < \sqrt{3}) = P(1 < X \leq \sqrt{3}).$$

Summary

1. A random variable is a real-valued function on the sample space Ω: $X : \Omega \to \mathbb{R}$.
2. From the inverse-mapping perspective, $\{X \leq x\} = X^{-1}((-\infty, x])$ is an event in the sample space; the random variable "pulls back" events on the real line to the sample space.
3. Three main benefits of introducing random variables: quantification, simplification of events, unified description of probability laws.
4. The distribution function $F(x) = P(X \leq x)$ completely describes the probability law of X.
5. The distribution function has three basic properties: monotonicity, boundedness, right-continuity.
6. To determine whether a function is a CDF, check all three properties one by one.
7. Through the degenerate, coin-toss, uniform, and mixed examples, we see that distribution functions come in many shapes: degenerate (single jump), discrete (step function), continuous (no jump), mixed (both continuous and discrete).

Exercises

1. Toss a fair coin three times. Let X be the number of heads. Write down all possible values of X and find its distribution function $F(x)$.
2. Roll two fair dice. Let X be the sum of the two numbers. Write down all possible values of X and find $P(X \leq 5)$.
3. Determine whether each of the following functions is a distribution function, and explain why:

   (a) $F(x) = \dfrac{1}{1+e^{-x}}$, $x \in \mathbb{R}$;

   (b) $F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < 1, \\ 1, & x \geq 1; \end{cases}$

   (c) $F(x) = \begin{cases} 0, & x < 0, \\ \sin x, & 0 \leq x < \pi, \\ 0, & x \geq \pi. \end{cases}$

Exercises (continued, Part 1)

4. Let the distribution function of X be

$$F(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \le x < 0, \\ 0.5, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Find $P(X = 0)$, $P(X < 0)$, $P(0 < X \le 1)$.

5. Let the distribution function of X be

$$F(x) = \begin{cases} 0, & x < 0, \\ Ax^2, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Find the constant A, and compute $P(0.5 < X \le 1)$.

Exercises (continued, Part 2)

6. Let the distribution function of X be

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{2} + \frac{x}{4}, & 0 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

Find $P(X < 1)$, $P(X = 1)$, $P(1 < X \le 2)$.

Thank you!